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United States Patent	12410133
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Huang; Lianfeng et al.

Crystalline (r)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, compositions and methods of use thereof

Abstract

Provided herein are crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are also disclosed.

Inventors: Huang; Lianfeng (Basking Ridge, NJ), Tsou; Nancy (Edison, NJ), White; Nicole Suzanne (San Diego, CA), Xu; Jun (Suzhou, CN), Zhang; Qun (Suzhou, CN)

Applicant: Celgene Corporation (Summit, NJ); Lundbeck La Jolla Research Center, Inc. (San Diego, CA)

Family ID: 68069873

Assignee: Celgene Corporation (Summit, NJ); Lundbeck La Jolla Research Center, Inc. (San Diego, CA)

Appl. No.: 18/143765

Filed: May 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230382861 A1	Nov. 30, 2023

Related U.S. Application Data

continuation parent-doc US 17338182 20210603 US 11680046 child-doc US 18143765
division parent-doc US 16569578 20190912 US 11053199 20210706 child-doc US 17338182
us-provisional-application US 62731014 20180913

Publication Classification

Int. Cl.: C07D211/94 (20060101); A61P35/00 (20060101)

U.S. Cl.:

CPC C07D211/94 (20130101); A61P35/00 (20180101); C07B2200/13 (20130101)

Field of Classification Search

CPC: C07D (211/94); A61P (35/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11053199	12/2020	Huang	N/A	A61P 35/00
11680046	12/2022	Huang et al.	N/A	N/A
2010/0009971	12/2009	Ishii et al.	N/A	N/A
2018/0256566	12/2017	Grice et al.	N/A	N/A
2019/0282568	12/2018	Grice et al.	N/A	N/A
2020/0095201	12/2019	Huang et al.	N/A	N/A
2021/0363105	12/2020	Huang et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 2018/169880	12/2017	WO	A61K 31/497

OTHER PUBLICATIONS

Bernstein, 2004, "Crystal Structure Prediction and Polymorphism," ACA Transactions, 39:14-23. cited by applicant

Braga et al., 2005, "Making crystals from crystals: a green route to crystal engineering and polymorphism," Chemical Communications, 3635-3645. cited by applicant

Chemburkar et al., 2000, "Dealing with the Impact of Ritonavir Polymorphson the Late Stages of Bulk Drug Process Development," Organic Process Research & Development, 4(5):413-417. cited by applicant

Di Martino et al., 1997, "Preparation and Physical Characterization of Forms II and III of Paracetamol," Journal of Thermal Analysis, 48(3):447-458. cited by applicant

International Search Report and Written Opinion dated Nov. 14, 2019 for PCT/US2019/050771 (9 pages). cited by applicant

Jones et al., 2011, "Pharmaceutical Cocrystals: An Emerging Approach to Physical Property Enhancement," MRS Bulletin, 31:875-879. cited by applicant

Knapman et al., 2000, "Polymorphic predictions," *Modern Drug Discovery*, 3:53-57. cited by applicant

Price, 2004, "The computational prediction of pharmaceutical crystal structures and polymorphism," *Advanced Drug Delivery Reviews*, 56(3):301-319. cited by applicant

Primary Examiner: Leeser; Erich A

Attorney, Agent or Firm: Jones Day

Background/Summary

CROSS-REFERENCE (1) This application is a continuation of U.S. application Ser. No. 17/338,182, filed Jun. 3, 2021, which is a division of U.S. application Ser. No. 16/569,578, filed Sep. 12, 2019, now U.S. Pat. No. 11,053,199, which claims priority to U.S. Provisional Application No. 62/731,014, filed Sep. 13, 2018, each of which is herein incorporated by reference in its entirety.

FIELD

(1) Provided herein are solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions comprising such solid forms and methods of use for treating, preventing, and managing various disorders are also provided herein.

BACKGROUND

(2) Many compounds can exist in different crystal forms, or polymorphs, which exhibit different physical, chemical, and spectroscopic properties. For example, certain polymorphs of a compound may be more readily soluble in particular solvents, may flow more readily, or may compress more easily than others. See, e.g., P. DiMartino, et al., *J. Thermal Anal.*, 48:447-458 (1997). In the case of drugs, certain solid forms may be more bioavailable than others, while others may be more stable under certain manufacturing, storage, and biological conditions.

(3) Polymorphic forms of a compound are known in the pharmaceutical arts to affect, for example, the solubility, stability, flowability, fractability, and compressibility of the compound, as well as the safety and efficacy of drug products comprising it. See, e.g., Knapman, K. *Modern Drug Discoveries*, 2000, 53. Therefore, the discovery of new polymorphs of a drug can provide a variety of advantages.

(4) The identification and selection of a solid form of a pharmaceutical compound are complex, given that a change in solid form may affect a variety of physical and chemical properties, which may provide benefits or drawbacks in processing, formulation, stability, bioavailability, storage, handling (e.g., shipping), among other important pharmaceutical characteristics. Useful pharmaceutical solids include crystalline solids and amorphous solids, depending on the product and its mode of administration. Amorphous solids are characterized by a lack of long-range structural order, whereas crystalline solids are characterized by structural periodicity. The desired class of pharmaceutical solid depends upon the specific application; amorphous solids are sometimes selected on the basis of, e.g., an enhanced dissolution profile, while crystalline solids may be desirable for properties such as, e.g., physical or chemical stability.

(5) The importance of discovering polymorphs was underscored by the case of Ritonavir™, an HIV protease inhibitor that was formulated as soft gelatin capsules. About two years after the product was launched, the unanticipated precipitation of a new, less soluble polymorph in the formulation necessitated the withdrawal of the product from the market until a more consistent formulation could be developed (see S. R. Chemburkar et al., *Org. Process Res. Dev.*, (2000) 4:413-417).

(6) Notably, it is not possible to predict a priori if crystalline forms of a compound even exist, let alone how to successfully prepare them (see, e.g., Braga and Grepioni, 2005, "Making crystals from crystals: a green route to crystal engineering and polymorphism," *Chem. Commun.* :3635-3645 (with respect to crystal engineering, if instructions are not very precise and/or if other external factors affect the process, the result can be unpredictable); Jones et al., 2006, Pharmaceutical Cocrystals: An Emerging Approach to Physical Property Enhancement," *MRS Bulletin* 31:875-879 (At present it is not generally possible to computationally predict the number of observable polymorphs of even the simplest molecules); Price, 2004, "The computational prediction of pharmaceutical crystal structures and polymorphism," *Advanced Drug Delivery Reviews* 56:301-319 ("Price"); and Bernstein, 2004, "Crystal Structure Prediction and Polymorphism," *ACA Transactions* 39:14-23 (a great deal still needs to be learned and done before one can state with any degree of confidence the ability to predict a crystal structure, much less polymorphic forms)).

(7) The variety of possible solid forms creates potential diversity in physical and chemical properties for a given pharmaceutical compound. The discovery and selection of solid forms are of great importance in the development of an effective, stable and marketable pharmaceutical product.

(8) (R)-5-Carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is a compound which is a dual MAGL and FAAH inhibitor, and may be used in treating various disorders.

(9) The synthesis of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate has been described in PCT/US2018/022049 filed Mar. 12, 2018.

(10) Novel crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are described herein. New polymorphic forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can further the development of formulations for the treatment of various illnesses, and may yield numerous formulation, manufacturing and therapeutic benefits.

SUMMARY

(11) Provided herein are crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Also provided herein are pharmaceutical compositions comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Further provided herein are methods of treating or preventing a variety of disease and disorders, which comprise administering to a patient a therapeutically effective amount of a crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

(12) Also provided herein are methods of preparing, isolating, and characterizing crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 provides a representative X-ray powder diffraction (XRPD) pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(2) FIG. 2 provides a representative differential scanning calorimetry (DSC) thermogram and a representative thermogravimetric analysis (TGA) thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(3) FIG. 3 provides a representative dynamic vapor sorption (DVS) plot of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(4) FIG. 4 provides representative XRPD patterns of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A before and after a DVS

experiment.

(5) FIG. 5 provides representative XRPD patterns of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A before and after accelerated conditions stability tests.

(6) FIG. 6 provides a representative nuclear magnetic resonance (NMR) spectrum of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(7) FIG. 7 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

(8) FIG. 8 provides a representative DSC thermogram and a representative TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

(9) FIG. 9 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

(10) FIG. 10 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

(11) FIG. 11 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

(12) FIG. 12 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

(13) FIG. 13 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

(14) FIG. 14 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

DEFINITIONS

(15) As used herein, and unless otherwise specified, the compound referred to herein as (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate corresponds to a compound of Formula (I), depicted below.

(16) ##STR00001##

(17) (R)-5-Carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained via standard, synthetic methods (see e.g., PCT/US2018/022049).

(18) Unless otherwise specified, the term “crystalline” and related terms used herein, when used to describe a substance, component, product, or form, mean that the substance, component, product, or form is substantially crystalline, for example, as determined by X-ray diffraction. (see, e.g., *Remington's Pharmaceutical Sciences*, 20^{sup}.th ed., Lippincott Williams & Wilkins, Philadelphia PA, 173 (2000); *The United States Pharmacopeia*, 37^{sup}.th ed., 503-509 (2014)).

(19) As used herein, and unless otherwise specified, the terms “about” and “approximately,” when used in connection with doses, amounts, or weight percents of ingredients of a composition or a dosage form, mean a dose, amount, or weight percent that is recognized by one of ordinary skill in the art to provide a pharmacological effect equivalent to that obtained from the specified dose, amount, or weight percent. In certain embodiments, the terms “about” and “approximately,” when used in this context, contemplate a dose, amount, or weight percent within 30%, within 20%, within 15%, within 10%, or within 5%, of the specified dose, amount, or weight percent.

(20) As used herein, and unless otherwise specified, the terms “about” and “approximately,” when used in connection with a numeric value or range of values which is provided to characterize a particular solid form, e.g., a specific temperature or temperature range, such as, for example, that describes a melting, dehydration, desolvation, or glass transition temperature; a mass change, such as, for example, a mass change as a function of temperature or humidity; a solvent or water content, in terms of, for example, mass or a percentage; or a peak position, such as, for example, in analysis by, for example, IR or Raman spectroscopy or XRPD; indicate that the value or range of values

may deviate to an extent deemed reasonable to one of ordinary skill in the art while still describing the solid form. Techniques for characterizing crystal forms and amorphous forms include, but are not limited to, thermal gravimetric analysis (TGA), differential scanning calorimetry (DSC), X-ray powder diffractometry (XRPD), single-crystal X-ray diffractometry, vibrational spectroscopy, e.g., infrared (IR) and Raman spectroscopy, solid-state and solution nuclear magnetic resonance (NMR) spectroscopy, optical microscopy, hot stage optical microscopy, scanning electron microscopy (SEM), electron crystallography and quantitative analysis, particle size analysis (PSA), surface area analysis, solubility studies, and dissolution studies. In certain embodiments, the terms “about” and “approximately,” when used in this context, indicate that the numeric value or range of values may vary within 30%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1.5%, 1%, 0.5%, or of the recited value or range of values. In the context of molar ratios, “about” and “approximately” indicate that the numeric value or range of values may vary within 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1.5%, 1%, 0.5%, or 0.25% of the recited value or range of values. It should be understood that the numerical values of the peaks of an X-ray powder diffraction pattern may vary from one machine to another, or from one sample to another, and so the values quoted are not to be construed as absolute, but with an allowable variability, such as ± 0.2 degrees two theta ($^{\circ} 2\theta$), or more. For example, in some embodiments, the value of an XRPD peak position may vary by up to ± 0.2 degrees 2θ while still describing the particular XRPD peak.

(21) As used herein, and unless otherwise specified, a solid form that is “substantially physically pure” is substantially free from other solid forms. In certain embodiments, a crystal form that is substantially physically pure contains less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, or 0.01% of one or more other solid forms on a weight basis. The detection of other solid forms can be accomplished by any method apparent to a person of ordinary skill in the art, including, but not limited to, diffraction analysis, thermal analysis, elemental combustion analysis and/or spectroscopic analysis.

(22) As used herein, and unless otherwise specified, a solid form that is “substantially chemically pure” is substantially free from other chemical compounds (i.e., chemical impurities). In certain embodiments, a solid form that is substantially chemically pure contains less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.01% of one or more other chemical compounds on a weight basis. The detection of other chemical compounds can be accomplished by any method apparent to a person of ordinary skill in the art, including, but not limited to, methods of chemical analysis, such as, e.g., mass spectrometry analysis, spectroscopic analysis, thermal analysis, elemental combustion analysis and/or chromatographic analysis.

(23) As used herein, and unless otherwise indicated, a chemical compound, solid form, or composition that is “substantially free” of another chemical compound, solid form, or composition means that the compound, solid form, or composition contains, in certain embodiments, less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.01% by weight of the other compound, solid form, or composition.

(24) Unless otherwise specified, the terms “anhydrate” and “anhydrous” as used herein, refer to a solid form of a substance which does not contain water in its crystal lattice.

(25) As used herein, and unless otherwise specified, the terms “treat,” “treating” and “treatment” refer to the eradication or amelioration of a disease or disorder, or of one or more symptoms associated with the disease or disorder. In certain embodiments, the terms refer to minimizing the spread or worsening of the disease or disorder resulting from the administration of one or more prophylactic or therapeutic agents to a subject with such a disease or disorder. In some embodiments, the terms refer to the administration of a compound provided herein, with or without other additional active agents, after the onset of symptoms of a particular disease.

(26) Unless otherwise specified, the term “composition” as used herein is intended to encompass a

product comprising the specified ingredient(s) (and in the specified amount(s), if indicated), as well as any product which results, directly or indirectly, from combination of the specified ingredient(s) in the specified amount(s). By “pharmaceutically acceptable,” it is meant a diluent, excipient, or carrier in a formulation must be compatible with the other ingredient(s) of the formulation and not deleterious to the recipient thereof.

(27) Unless otherwise specified, the term “subject” is defined herein to include animals, such as mammals, including, but not limited to, primates (e.g., humans), cows, sheep, goats, horses, dogs, cats, rabbits, rats, mice, and the like. In specific embodiments, the subject is a human.

(28) Unless otherwise specified, to the extent that there is a discrepancy between a depicted chemical structure of a compound provided herein and a chemical name of a compound provided herein, the chemical structure shall control.

DETAILED DESCRIPTION

(29) Provided herein are solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. In certain embodiments, the solid form is crystalline. In certain embodiments, the solid form is a single-component solid form. In certain embodiments, the solid form is an anhydrate.

(30) Crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be prepared by the methods described herein, including the methods described in the Examples below, or by techniques known in the art, including heating, cooling, freeze drying, lyophilization, solvent evaporation, solvent recrystallization, antisolvent addition, slurry recrystallization, rapid cooling, slow cooling, exposure to solvent and/or water, and drying. The particle size of the resulting solid forms, which can vary, e.g., from nanometer dimensions to millimeter dimensions, can be controlled, e.g., by varying crystallization conditions, such as, e.g., the rate of crystallization and/or the crystallization solvent system, or by particle-size reduction techniques, e.g., grinding, milling, micronizing or sonication.

(31) While not intending to be bound by any particular theory, crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are characterized by physical properties, e.g., stability, solubility and dissolution rate, appropriate for pharmaceutical and therapeutic dosage forms. Moreover, while not wishing to be bound by any particular theory, crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are characterized by physical properties (e.g., density, compressibility, hardness, morphology, cleavage, stickiness, solubility, water uptake, electrical properties, thermal behavior, solid-state reactivity, physical stability, and chemical stability) affecting particular processes (e.g., yield, filtration, washing, drying, milling, mixing, tableting, flowability, dissolution, formulation, and lyophilization) which make certain crystalline forms suitable for the manufacture of a dosage form. Such properties can be determined using particular analytical chemical techniques, including solid-state analytical techniques (e.g., X-ray diffraction, microscopy, spectroscopy and thermal analysis), as described herein and known in the art.

(32) Certain embodiments herein provide compositions comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

(33) Certain embodiments provide methods of using these compositions in the treatment, prevention or management of diseases and disorders including, but not limited to, diseases and disorders related to MAGL and/or FAAH activity.

(34) Certain embodiments herein provide methods of using crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate in the treatment, prevention, or management of pain, neurological disorders, anxiety, inflammatory bowel disease, neuropathic pain, and the proliferation and migration of cancer cells.

(35) Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(36) In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from methanol. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from isopropyl acetate. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from a 9:1 ethanol/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from toluene.

(37) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A may be characterized by X-ray powder diffraction analysis.

(38) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 16.5, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ .

(39) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 16.4, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ .

(40) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 18.1, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ .

(41) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.6 degrees $2\theta \pm 0.2$ degrees 2θ .

(42) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.7 degrees $2\theta \pm 0.2$ degrees 2θ .

(43) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, or fourteen of the following approximate positions: 9.9, 13.3, 13.9, 14.8, 16.4, 18.0, 19.9, 20.1, 22.3, 23.0, 24.4, 24.6, 26.9, and 29.3 degrees 2θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

(44) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty, or twenty-one of the following approximate positions: 9.9, 13.4, 14.0, 14.2, 14.9, 16.0, 16.5, 17.3, 18.1, 19.3, 19.9, 20.1, 20.6, 21.4, 22.3, 23.1, 24.1, 24.7, 27.0, 27.2, and 29.3 degrees 2θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

(45) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction pattern depicted in FIG. 1.

(46) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A may be characterized by thermal

analysis.

(47) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 113° C.

(48) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 112° C.

(49) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 112° C.

(50) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 111° C.

(51) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 2.

(52) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is anhydrous.

(53) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.22% when heated from about 25° C. to about 100° C.

(54) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

(55) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

(56) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 2.

(57) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is substantially physically pure.

(58) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is substantially chemically pure.

(59) In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

(60) Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

(61) In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B can be obtained from a 9:1 acetonitrile/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B can be obtained from an ethyl acetate/n-heptane mixture, methyl acetate, acetone or methylene chloride.

(62) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-

(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B may be characterized by X-ray powder diffraction analysis.

(63) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees $2\theta \pm 0.2$ degrees 2θ .

(64) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B wherein the X-ray powder diffraction pattern further comprises peaks at 15.1, 18.7, and 20.1 degrees $2\theta \pm 0.2$ degrees 2θ .

(65) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve of the following approximate positions: 15.1, 16.2, 17.9, 18.2, 18.7, 19.2, 20.1, 21.1, 24.3, 25.0, 27.4, and 27.6 degrees 2θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

(66) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 7.

(67) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B may be characterized by thermal analysis.

(68) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 95° C.

(69) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 91° C.

(70) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 8.

(71) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is anhydrous.

(72) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 2.5% when heated from about 30° C. to about 100° C.

(73) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

(74) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

(75) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 8.

(76) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is substantially physically pure.

(77) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is substantially chemically pure.

(78) In one embodiment, provided is a pharmaceutical composition comprising a crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

(79) Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

(80) In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C can be obtained from a 1:3 chloroform/hexane mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from a 1:1 ethanol/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from a 3:1 acetonitrile/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from slurring Form B in FaSSIF. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from slurring Form B in FeSSIF.

(81) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C may be characterized by X-ray powder diffraction analysis.

(82) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees $2\theta \pm 0.2$ degrees 2θ .

(83) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C wherein the X-ray powder diffraction pattern further comprises peaks at 12.3, 16.8, and 25.5 degrees $2\theta \pm 0.2$ degrees 2θ .

(84) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty, twenty-one, or twenty-two of the following approximate positions: 12.3, 14.1, 14.7, 14.8, 15.5, 15.7, 16.2, 16.8, 17.1, 18.0, 18.2, 18.5, 19.5, 20.0, 20.5, 21.0, 21.4, 22.8, 24.6, 25.5, 25.9, and 29.6 degrees 2θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

(85) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 9.

(86) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C may be characterized by thermal analysis.

(87) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 99° C.

(88) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 98° C.

- (89) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermograms depicted in FIG. 10.
- (90) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is anhydrous.
- (91) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of about 0.149% when heated from about 30° C. to about 150° C.
- (92) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.
- (93) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of about 0.15% when heated from about 30° C. to about 150° C.
- (94) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 10.
- (95) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is substantially physically pure.
- (96) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is substantially chemically pure.
- (97) In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.
- (98) Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.
- (99) In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D can be obtained from slurrying Form B in methyl tert-butyl ether (MTBE).
- (100) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D may be characterized by X-ray powder diffraction analysis.
- (101) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees 2 θ $\pm$ 0.2 degrees 2 θ .
- (102) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D wherein the X-ray powder diffraction pattern further comprises peaks at 18.3, 26.7, and 29.5 degrees 2 θ $\pm$ 0.2 degrees 2 θ .
- (103) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, or seventeen of the following approximate positions: 13.4, 14.0, 14.8, 16.1, 16.8, 18.0, 18.3, 18.6, 19.5, 19.8, 20.7, 21.0, 22.8, 23.8, 24.4, 26.7, and 29.5 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks

matching peaks in the representative XRPD pattern provided herein.

(104) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 11.

(105) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D may be characterized by thermal analysis.

(106) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 101° C.

(107) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 99° C.

(108) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 12.

(109) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is anhydrous.

(110) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram comprising a weight loss of about 0.054% when heated from about 30° C. to about 150° C.

(111) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

(112) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 12.

(113) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is substantially physically pure.

(114) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is substantially chemically pure.

(115) In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

(116) Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

(117) In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E can be obtained from a chloroform/hexane mixture.

(118) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E may be characterized by X-ray powder diffraction analysis.

(119) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees $2\theta \pm 0.2$ degrees 2θ .

(120) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E wherein the X-ray powder diffraction

pattern further comprises peaks at 15.8, 18.4, and 21.9 degrees $2\theta \pm 0.2$ degrees 2θ .

(121) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve of the following approximate positions: 10.9, 12.7, 15.8, 17.2, 17.8, 18.4, 19.7, 21.3, 21.9, 22.8, 24.2, and 25.6 degrees 2θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

(122) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 13.

(123) In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E may be characterized by thermal analysis.

(124) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 104° C.

(125) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 102° C.

(126) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermograms depicted in FIG. 14.

(127) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is anhydrous.

(128) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric analysis thermogram comprising a weight loss of about 0.37% when heated from about 30° C. to about 150° C.

(129) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

(130) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 14.

(131) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is substantially physically pure.

(132) In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is substantially chemically pure.

(133) In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

(134) Pharmaceutical Compositions

(135) Pharmaceutical compositions and single unit dosage forms comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are provided herein. Also provided herein are methods for preparing pharmaceutical compositions and single unit dosage forms comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-

(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. For example, in certain embodiments, individual dosage forms comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate provided herein or prepared using crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate provided herein may be suitable for oral, mucosal (including rectal, nasal, or vaginal), parenteral (including subcutaneous, intramuscular, bolus injection, intraarterial, or intravenous), sublingual, transdermal, buccal, or topical administration.

(136) In certain embodiments, pharmaceutical compositions and dosage forms provided herein comprise crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Certain embodiments herein provide pharmaceutical compositions and dosage forms comprising crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, wherein the crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is substantially pure. Certain embodiments herein provide pharmaceutical compositions and dosage forms comprising crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate as provided herein, which is substantially free of other crystalline solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate and/or amorphous solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions and dosage forms provided herein typically also comprise one or more pharmaceutically acceptable excipients, diluents or carriers.

(137) Single unit dosage forms provided herein are suitable for oral or parenteral (e.g., subcutaneous, intravenous, bolus injection, intramuscular, or intraarterial) administration to a patient. Examples of dosage forms include, but are not limited to: tablets; caplets; capsules, such as soft elastic gelatin capsules; powders and sterile solids that can be reconstituted to provide liquid dosage forms suitable for parenteral administration to a patient.

(138) The composition, shape, and type of dosage forms provided herein will typically vary depending on their use. A parenteral dosage form may contain smaller amounts of one or more of the active ingredients it comprises than an oral dosage form used to treat the same disease or disorder. These and other ways in which specific dosage forms encompassed by this invention will vary from one another will be readily apparent to those skilled in the art. See, e.g., *Remington's Pharmaceutical Sciences*, 18th ed., Mack Publishing, Easton Pa. (1990).

(139) Typical pharmaceutical compositions and dosage forms comprise one or more excipients. Suitable excipients are well known to those skilled in the art of pharmacy, and non-limiting examples of suitable excipients are provided herein. Whether a particular excipient is suitable for incorporation into a pharmaceutical composition or dosage form depends on a variety of factors well known in the art including, but not limited to, the way in which the dosage form will be administered to a patient. For example, oral dosage forms such as tablets may contain excipients not suited for use in parenteral dosage forms. The suitability of a particular excipient may also depend on the specific active ingredients in the dosage form.

(140) Like the amounts and types of excipients, the amounts and specific types of active ingredients in a dosage form may differ depending on factors such as, but not limited to, the route by which it is to be administered to patients. However, typical dosage forms provided herein lie within the range of from about 0.1 mg to about 1,000 mg per day, given as a single once-a-day dose in the morning or as divided doses throughout the day. More specifically, the daily dose may be administered twice, three times, or four times daily in equally divided doses. Specifically, a daily dose range may be from about 0.1 mg to about 500 mg per day, more specifically, between about 0.1 mg and about 200 mg per day. A daily dose range may be 1 mg, 2 mg, 3 mg, 4 mg, or 5 mg. In managing the patient, the therapy may be initiated at a lower dose, perhaps about 1 mg to about 25

mg, and increased if necessary up to about 200 mg to about 1,000 mg per day as either a single dose or divided doses, depending on the patient's global response.

(141) Oral Dosage Forms

(142) Pharmaceutical compositions provided herein that are suitable for oral administration can be presented as discrete dosage forms, such as, but are not limited to, tablets (e.g., chewable tablets), caplets, capsules, and liquids (e.g., flavored syrups). Such dosage forms contain predetermined amounts of active ingredients, and may be prepared by methods of pharmacy well known to those skilled in the art. See generally Remington's Pharmaceutical Sciences, 18th ed., Mack Publishing, Easton Pa. (1990).

(143) Typical oral dosage forms provided herein are prepared by combining the active ingredient(s) in an intimate admixture with at least one excipient according to conventional pharmaceutical compounding techniques. Excipients can take a wide variety of forms depending on the form of preparation desired for administration. For example, excipients suitable for use in oral liquid or aerosol dosage forms include, but are not limited to, water, glycols, oils, alcohols, flavoring agents, preservatives, and coloring agents. Examples of excipients suitable for use in solid oral dosage forms (e.g., powders, tablets, capsules, and caplets) include, but are not limited to, starches, sugars, micro-crystalline cellulose, diluents, granulating agents, lubricants, binders, and disintegrating agents.

(144) If desired, tablets can be coated by standard aqueous or nonaqueous techniques. Such dosage forms can be prepared by any of the methods of pharmacy. In general, pharmaceutical compositions and dosage forms are prepared by uniformly and intimately admixing the active ingredients with liquid carriers, finely divided solid carriers, or both, and then shaping the product into the desired presentation if necessary.

(145) For example, a tablet can be prepared by compression or molding. Compressed tablets can be prepared by compressing in a suitable machine the active ingredients in a free-flowing form such as powder or granules, optionally mixed with an excipient. Molded tablets can be made by molding in a suitable machine a mixture of the powdered compound moistened with an inert liquid diluent.

(146) Examples of excipients that can be used in oral dosage forms of the invention include, but are not limited to, binders, fillers, disintegrants, and lubricants. Binders suitable for use in pharmaceutical compositions and dosage forms include, but are not limited to, corn starch, potato starch, or other starches, gelatin, natural and synthetic gums such as acacia, sodium alginate, alginic acid, other alginates, powdered tragacanth, guar gum, cellulose and its derivatives (e.g., ethyl cellulose, cellulose acetate, carboxymethyl cellulose calcium, sodium carboxymethyl cellulose), polyvinyl pyrrolidone, methyl cellulose, pre-gelatinized starch, hydroxypropyl methyl cellulose, (e.g., Nos. 2208, 2906, 2910), microcrystalline cellulose, and mixtures thereof.

(147) Examples of fillers suitable for use in the pharmaceutical compositions and dosage forms disclosed herein include, but are not limited to, talc, calcium carbonate (e.g., granules or powder), microcrystalline cellulose, powdered cellulose, dextrates, kaolin, mannitol, silicic acid, sorbitol, starch, pre-gelatinized starch, and mixtures thereof. The binder or filler in pharmaceutical compositions of the invention is typically present in from about 50 to about 99 weight percent of the pharmaceutical composition or dosage form.

(148) Suitable forms of microcrystalline cellulose include, but are not limited to, the materials sold as AVICEL-PH-101™, AVICEL-PH-103™, AVICEL RC-581™, AVICEL-PH-105™ (available from FMC Corporation, American Viscose Division, Avicel Sales, Marcus Hook, Pa.), and mixtures thereof. A specific binder is a mixture of microcrystalline cellulose and sodium carboxymethyl cellulose sold as AVICEL RC-58™. Suitable anhydrous or low moisture excipients or additives include AVICEL-PH-103™ and Starch 1500 LM™.

(149) Disintegrants are used in the compositions provided herein to provide tablets that disintegrate when exposed to an aqueous environment. Tablets that contain too much disintegrant may disintegrate in storage, while those that contain too little may not disintegrate at a desired rate or

under the desired conditions. Thus, a sufficient amount of disintegrant that is neither too much nor too little to detrimentally alter the release of the active ingredients should be used to form solid oral dosage forms of the invention. The amount of disintegrant used varies based upon the type of formulation, and is readily discernible to those of ordinary skill in the art. Typical pharmaceutical compositions comprise from about 0.5 to about 15 weight percent of disintegrant, specifically from about 1 to about 5 weight percent of disintegrant.

(150) Disintegrants that can be used in pharmaceutical compositions and dosage forms provided herein include, but are not limited to, agar-agar, alginic acid, calcium carbonate, microcrystalline cellulose, croscarmellose sodium, crospovidone, polacrillin potassium, sodium starch glycolate, potato or tapioca starch, pre-gelatinized starch, other starches, clays, other algins, other celluloses, gums, and mixtures thereof

(151) Lubricants that can be used in pharmaceutical compositions and dosage forms provided herein include, but are not limited to, calcium stearate, magnesium stearate, mineral oil, light mineral oil, glycerin, sorbitol, mannitol, polyethylene glycol, other glycols, stearic acid, sodium lauryl sulfate, talc, hydrogenated vegetable oil (e.g., peanut oil, cottonseed oil, sunflower oil, sesame oil, olive oil, corn oil, and soybean oil), zinc stearate, ethyl oleate, ethyl laureate, agar, and mixtures thereof. Additional lubricants include, for example, a syloid silica gel (AEROSIL 200™, manufactured by W.R. Grace Co. of Baltimore, Md.), a coagulated aerosol of synthetic silica (marketed by Degussa Co. of Plano, Tex.), CAB-O-SIL™ (a pyrogenic silicon dioxide product sold by Cabot Co. of Boston, Mass.), and mixtures thereof. If used at all, lubricants are typically used in an amount of less than about one weight percent of the pharmaceutical compositions or dosage forms into which they are incorporated.

(152) Parenteral Dosage Forms

(153) Parenteral dosage forms can be administered to patients by various routes including, but not limited to, subcutaneous, intravenous (including bolus injection), intramuscular, and intraarterial injection. Because their administration typically bypasses patients' natural defenses against contaminants, parenteral dosage forms are preferably sterile or capable of being sterilized prior to administration to a patient. Examples of parenteral dosage forms include, but are not limited to, solutions ready for injection, dry products ready to be dissolved or suspended in a pharmaceutically acceptable vehicle for injection, suspensions ready for injection, and emulsions.

(154) Suitable vehicles that can be used to provide parenteral dosage forms provided herein are well known to those skilled in the art. Examples include, but are not limited to: Water for Injection USP; aqueous vehicles such as, but not limited to, Sodium Chloride Injection, Ringer's Injection, Dextrose Injection, Dextrose and Sodium Chloride Injection, and Lactated Ringer's Injection; water-miscible vehicles such as, but not limited to, ethyl alcohol, polyethylene glycol, and polypropylene glycol; and non-aqueous vehicles such as, but not limited to, corn oil, cottonseed oil, peanut oil, sesame oil, ethyl oleate, isopropyl myristate, and benzyl benzoate.

(155) Compounds that increase the solubility of one or more of the active ingredients disclosed herein can also be incorporated into the parenteral dosage forms provided herein.

(156) Combination Treatment

(157) Also contemplated herein are combination therapies, for example, co-administering a disclosed compound and an additional active agent, as part of a specific treatment regimen intended to provide the beneficial effect from the co-action of these therapeutic agents. The beneficial effect of the combination includes, but is not limited to, pharmacokinetic or pharmacodynamic co-action resulting from the combination of therapeutic agents. Administration of these therapeutic agents in combination typically is carried out over a defined time period (usually weeks, months, or years depending upon the combination selected). Combination therapy is intended to embrace administration of multiple therapeutic agents in a sequential manner, that is, wherein each therapeutic agent is administered at a different time, as well as administration of these therapeutic agents, or at least two of the therapeutic agents, in a substantially simultaneous manner.

(158) Substantially simultaneous administration is accomplished, for example, by administering to the subject a single formulation or composition, (e.g., a tablet or capsule having a fixed ratio of each therapeutic agent or in multiple, single formulations (e.g., capsules) for each of the therapeutic agents. Sequential or substantially simultaneous administration of each therapeutic agent is effected by any appropriate route including, but not limited to, oral routes, intravenous routes, intramuscular routes, and direct absorption through mucous membrane tissues. The therapeutic agents are administered by the same route or by different routes. For example, a first therapeutic agent of the combination selected is administered by intravenous injection while the other therapeutic agents of the combination are administered orally. Alternatively, for example, all therapeutic agents are administered orally or all therapeutic agents are administered by intravenous injection.

(159) Combination therapy also embraces the administration of the therapeutic agents as described above in further combination with other biologically active ingredients and non-drug therapies. Where the combination therapy further comprises a non-drug treatment, the non-drug treatment is conducted at any suitable time so long as a beneficial effect from the co-action of the combination of the therapeutic agents and non-drug treatment is achieved. For example, in appropriate cases, the beneficial effect is still achieved when the non-drug treatment is temporally removed from the administration of the therapeutic agents, perhaps by days or even weeks.

(160) The components of the combination are administered to a patient simultaneously or sequentially. It will be appreciated that the components are present in the same pharmaceutically acceptable carrier and, therefore, are administered simultaneously. Alternatively, the active ingredients are present in separate pharmaceutical carriers, such as conventional oral dosage forms, that are administered either simultaneously or sequentially.

(161) For example, e.g., for contemplated treatment of pain, a disclosed compound is co-administered with another therapeutic for pain such as an opioid, a cannabinoid receptor (CB-1 or CB-2) modulator, a COX-2 inhibitor, acetaminophen, and/or a non-steroidal anti-inflammatory agent. Additional therapeutics e.g., for the treatment of pain that are co-administered, include morphine, codeine, hydromorphone, hydrocodone, oxymorphone, fentanyl, tramadol, and levorphanol.

(162) Other contemplated therapeutics for co-administration include aspirin, naproxen, ibuprofen, salsalate, diflunisal, dexibuprofen, fenoprofen, ketoprofen, oxaprozin, loxoprofen, indomethacin, tolmetin, sulindac, etodolac, ketorolac, piroxicam, meloxicam, tenoxicam, droxicam, lornoxicam, celecoxib, parecoxib, rimonabant, and/or etoricoxib.

EXAMPLES

Example 1: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A

(163) Form A was obtained with several solvents, including methanol, isopropyl acetate, and a 9:1 mixture of ethanol/water. Form A was also prepared by slurring starting material (a mixture of Form A/C) in toluene at RT for 1 day. NMR spectra of Form A indicates trace amount of surface toluene. HPLC purity of Form A sample was measured to be 99.87 area%. Form A has a differential scanning calorimetry thermogram comprising an endotherm with an onset of about 111-112° C. and a maximum at about 113° C. (shown in FIG. 2). Form A has a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 100° C., indicating anhydrous material (shown in FIG. 2).

(164) The dynamic vapor sorption (DVS) isotherm plot of Form A was collected at 25° C. between 0 and 95% RH (shown in FIG. 3). A water uptake of 0.05% was observed at 25° C./80% RH. The XRPD comparison indicated that no form change was observed for Form A after the DVS test (shown in FIG. 4).

(165) Form A sample showed good physical and chemical stability under accelerated conditions of 25° C./60% RH and 40° C./75% RH for 3 months. X-ray powder diffractograms comparing Form A before and after accelerated conditions is shown in FIG. 5. XRPD peaks for Form A are shown in

Tables 1 and 2 (different XRPD instruments and samples; peaks below a certain amplitude are omitted for clarity).

(166) TABLE-US-00001 TABLE 1 XRPD peaks of Form A. No. 2-theta(deg) d(ang.)

Height(counts) 1 4.9 17.85 445 2 9.9 8.906 7062 3 12.5 7.086 450 4 13.4 6.625 1881 5 14.0 6.324 3185 6 14.2 6.212 3613 7 14.9 5.934 4377 8 16.0 5.545 1117 9 16.5 5.378 3668 10 16.7 5.302 461 11 17.3 5.131 2862 12 18.1 4.888 10138 13 18.8 4.721 543 14 19.3 4.589 2757 15 19.7 4.50 561 16 19.9 4.449 6638 17 20.1 4.405 12550 18 20.6 4.311 3593 19 21.4 4.151 2679 20 22.3 3.975 21984 21 23.1 3.851 3116 22 24.1 3.684 1875 23 24.7 3.605 4728 24 25.1 3.551 636 25 26.1 3.415 947 26 27.0 3.301 2208 27 27.2 3.279 2457 28 28.0 3.185 852 29 28.8 3.100 535 30 29.3 3.041 1477 31 31.2 2.863 536 32 32.2 2.777 594 33 34.3 2.615 457 34 36.7 2.446 592 35 38.2 2.352 607

(167) TABLE-US-00002 TABLE 2 XRPD peaks of Form A. 2- No. theta(deg) d(ang.)

Height(counts) 1 4.9 17.85 445 2 9.9 8.906 7062 3 12.5 7.086 450 4 13.4 6.625 1881 5 14.0 6.324 3185 6 14.2 6.212 3613 7 14.9 5.934 4377 8 16.0 5.545 1117 9 16.5 5.378 3668 10 16.7 5.302 461 11 17.3 5.131 2862 12 18.1 4.888 10138 13 18.8 4.721 543 14 19.3 4.589 2757 15 19.7 4.50 561 16 19.9 4.449 6638 17 20.1 4.405 12550 18 20.6 4.311 3593 19 21.4 4.151 2679 20 22.3 3.975 21984 21 23.1 3.851 3116 22 24.1 3.684 1875 23 24.7 3.605 4728 24 25.1 3.551 636 25 26.1 3.415 947 26 27.0 3.301 2208 27 27.2 3.279 2457 28 28.0 3.185 852 29 28.8 3.100 535 30 29.3 3.041 1477 31 31.2 2.863 536 32 32.2 2.777 594 33 34.3 2.615 457 34 36.7 2.446 592 35 38.2 2.352 607

Example 2: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B

(168) Form B was obtained with a 9:1 mixture of acetonitrile/water. Form B was also obtained via liquid vapor diffusion from mixture of ethyl acetate/n-heptane as well as evaporation in neat methyl acetate, acetone, or methylene chloride. The XRPD peaks for Form B are displayed in Table 3. A weight loss of less than 1.0% was observed when heated from about 30° C. to about 100° C. The DSC result exhibited a sharp endotherm onset at 91° C. and an endotherm with a maximum at about 95° C. Based on the low TGA weight loss and the single sharp DSC endotherm, Form B was postulated to be an anhydrate.

(169) TABLE-US-00003 TABLE 3 XRPD peaks of Form B. 2- Height No. theta(deg) d(ang.)

(counts) 1 6.9 12.86 341 2 9.1 9.74 232 3 10.4 8.522 563 4 11.5 7.700 142 5 12.7 6.985 359 6 13.1 6.730 257 7 13.8 6.39 273 8 14.3 6.182 473 9 15.1 5.847 2206 10 16.2 5.465 8503 11 17.3 5.114 551 12 17.9 4.950 2243 13 18.2 4.871 9600 14 18.7 4.754 4314 15 19.2 4.619 4926 16 20.1 4.421 3639 17 20.7 4.289 859 18 20.8 4.262 1649 19 21.1 4.205 2422 20 21.3 4.165 1181 21 21.6 4.120 421 22 22.7 3.922 1134 23 23.0 3.861 720 24 24.3 3.661 1424 25 25.0 3.566 2080 26 25.4 3.498 525 27 25.8 3.458 460 28 27.0 3.301 568 29 27.4 3.252 1417 30 27.6 3.224 827 31 28.4 3.143 238 32 29.4 3.036 402 33 29.9 2.991 128 34 30.6 2.924 715 35 31.7 2.823 129 36 33.5 2.673 139 37 34.8 2.575 198 38 37.0 2.431 179 39 39.5 2.282 190

Example 3: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C

(170) Form C was obtained when Form B was slurried in FaSSIF-v2 or FeSSIF-v2 at 37° C. for 2 hr. Form C can also be obtained with a 1:3 chloroform/hexane mixture. The DSC shows an endotherm with a maximum at about 99° C., and the TGA shows a weight loss of less than about 0.15% when heated from about 30° C. to about 150° C., indicating Form C is an anhydrate. XRPD peaks for Form C are shown in Table 4.

(171) TABLE-US-00004 TABLE 4 XRPD peaks of Form C. 2- Height No. theta(deg) d(ang.)

(counts) 1 5.1 17.18 1235 2 8.5 10.42 146 3 10.2 8.633 661 4 10.4 8.463 402 5 10.6 8.340 697 6 10.8 8.179 426 7 12.3 7.209 4781 8 12.6 7.007 943 9 14.1 6.294 1700 10 14.7 6.021 1871 11 14.8 5.963 2174 12 15.4 5.747 691 13 15.5 5.699 2577 14 15.7 5.628 1504 15 16.2 5.452 2030 16 16.8 5.259 5422 17 17.1 5.194 1475 18 17.3 5.135 611 19 18.0 4.928 9698 20 18.2 4.876 4834 21 18.5 4.798 6212 22 18.7 4.752 576 23 19.5 4.551 6357 24 20.0 4.444 11680 25 20.5 4.333 2281 26 21.0 4.223 15490 27 21.4 4.144 6065 28 21.8 4.070 1064 29 22.2 3.994 1216 30 22.8 3.893 4382 31

23.3 3.808 2959 32 23.8 3.736 989 33 24.6 3.616 553 34 25.5 3.490 6853 35 25.9 3.434 3080 36
26.4 3.375 372 37 26.6 3.348 605 38 27.0 3.305 1594 39 27.6 3.226 229 40 28.0 3.186 201 41 28.2
3.159 248 42 29.6 3.019 2177 43 31.2 2.868 498 44 31.3 2.852 398 45 32.0 2.797 477 46 32.8
2.728 299 47 33.3 2.692 205 48 34.2 2.621 165 49 34.5 2.598 269 50 36.0 2.490 417 51 36.5 2.462
472 52 37.2 2.418 92

Example 4: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D

(172) Form D can be obtained from slurrying Form B in MTBE at room temperature for 10 days. The DSC shows an endotherm with a maximum at about 101° C., and the TGA shows a weight loss of about 0.054% when heated from about 30° C. to about 150° C., indicating Form D is an anhydrate. XRPD peaks for Form D are shown in Table 5.

(173) TABLE-US-00005 TABLE 5 XRPD peaks of Form D. 2- Height No. theta(deg) d(ang.)
(counts) 1 5.9 15.094 2047 2 8.4 10.530 900 3 8.8 10.088 630 4 9.5 9.323 538 5 11.7 7.558 498 6
13.4 6.625 8528 7 14.0 6.305 2653 8 14.3 6.183 716 9 14.8 5.985 2408 10 15.3 5.794 365 11 16.1
5.510 4141 12 16.8 5.284 2986 13 17.8 4.981 907 14 18.0 4.924 1322 15 18.3 4.831 6368 16 18.6
4.762 2087 17 18.8 4.707 634 18 19.5 4.553 2557 19 19.8 4.488 2428 20 20.7 4.291 4145 21 21.0
4.218 11450 22 22.0 4.033 1042 23 22.8 3.893 1266 24 23.0 3.859 1217 25 23.8 3.734 23683 26
24.4 3.643 3299 27 24.8 3.588 1917 28 25.5 3.491 514 29 26.3 3.389 1610 30 26.7 3.341 8142 31
26.9 3.313 1471 32 28.0 3.187 402 33 29.5 3.021 5758 34 32.5 2.755 965 35 33.3 2.688 774 36
34.1 2.630 168 37 34.5 2.597 409 38 34.8 2.577 418 39 35.0 2.560 253 40 36.2 2.480 555 41 36.9
2.433 224 42 37.2 2.413 319 43 37.8 2.376 248 44 38.6 2.329 294 45 39.6 2.276 359

Example 5: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E

(174) Form E can be obtained from slurring Form C in a chloroform/hexane mixture at rt for 9 days. The DSC shows an endotherm with a maximum at about 104° C., and the TGA shows a weight loss of about 0.37% when heated from about 30° C. to about 150° C., indicating Form E is an anhydrate. XRPD peaks for Form E are shown in Table 6.

(175) TABLE-US-00006 TABLE 6 XRPD peaks for Form E. 2- Height No. theta(deg) d(ang.)
(counts) 1 7.5 11.73 333 2 9.1 9.71 115 3 10.9 8.144 698 4 12.7 7.006 1164 5 14.0 6.327 793 6 14.3
6.173 1113 7 15.8 5.622 2425 8 16.2 5.462 915 9 16.8 5.279 274 10 17.2 5.158 1845 11 17.8 4.985
2556 12 18.4 4.818 1879 13 19.7 4.502 1215 14 20.3 4.371 778 15 21.3 4.167 2749 16 21.9 4.054
2304 17 22.7 3.923 377 18 22.8 3.892 5540 19 23.5 3.790 358 20 24.2 3.681 1614 21 25.4 3.511
392 22 25.6 3.475 839 23 26.2 3.399 398 24 27.3 3.264 171 25 28.3 3.156 237 26 29.0 3.075 342
27 29.7 3.007 278 28 30.0 2.975 310 29 30.7 2.910 151

Example 6: Stability and Solubility Tests

(176) A solution stability test of Form A was performed to assess substantial degradation of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate in physiologically relevant media. A solution of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate (1 mg/mL) was prepared in acetonitrile/water (50/50 v/v). Test medium (1990 µL) was added to a vial. Test media evaluated included buffers pH 1-8, simulated gastric fluid (SGF), fasted state simulated intestinal fluid (FaSSIF), and fed state simulated intestinal fluid (FeSSIF). Stock solution (10 µL) was added to the vial containing the medium. This mixture was stirred to mix well and transferred to a HPLC vial, which was placed in the autosampler maintained at 37° C. The vial was injected repeatedly every 4 hr for 24 hr. Results are shown in Table 7. Solubility is shown in Table 8.

(177) TABLE-US-00007 TABLE 7 Solution chemical stability data for crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl- 4-(3-(trifluoromethoxy)benzyl) piperazine-1-carboxylate Form A at 37° C. Time (hr) pH = 1 pH = 3 pH = 5.0 pH = 6.8 pH = 8.0 SGF FaSSIF FeSSIF 0 100.0 100.0
100.0 100.0 100.0 100.0 100.0 100.0 4 99.6 99.8 100.9 100.2 101.3 100.2 100.6 99.6 8 100.3 99.8
101.7 100.8 100.5 100.1 100.2 100.2 12 100.3 98.5 99.9 100.1 99.7 99.2 100.5 99.7 16 99.9 99.3

100.7 100.0 100.1 99.7 100.3 99.6 20 101.9 99.6 99.9 99.6 98.7 100.4 100.0 99.8 24 100.5 99.5
99.6 99.0 99.4 99.6 100.0 99.7

(178) TABLE-US-00008 TABLE 8 Solubility of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A in physiologically relevant media (37° C.). Final Measured Medium pH Solubility (mg/mL)* pH 1.0 HCl/NaCl buffer 1.47 >20 pH 2.5 Phosphate buffer 3.44 4.326 pH 3 Phosphate buffer 4.20 1.251 pH 4 Acetate buffer 4.18 0.696 pH 5 Acetate buffer 5.14 0.091 pH 6.8 Phosphate buffer 6.88 0.007 pH 8.0 Phosphate buffer 8.10 0.005 SGF with pepsin 1.75 >20 FaSSIF 6.48 0.146 FeSSIF 5.17 1.018 pH 1 (0.1N HCl) 1.48 >20 pH 2 (0.01N HCl) 3.55 3.858 pH 3 (0.001N HCl) 4.90 0.057 pH 4 (0.0001N HCl) 7.78 0.011 Water 8.10 0.012

Example 7: Stress Polymorph Screen

(179) A stress polymorph screen using Form B as the starting form was performed. The tests were performed at room temperature or lower in a variety of solvents. Results are shown in Table 9.

(180) TABLE-US-00009 TABLE 9 Stress Polymorph Screen Using Form B as the Starting Form at room temperature or lower. Experiment Solvent Temp. State Result 1 Ethyl Acetate Room* Slurry Form A 2 Isopropyl Acetate Room Slurry Form A 3 Acetone 4° C. Evaporation Form B 4 Dichloromethane 4° C. Evaporation Form B 5 Acetonitrile Room* Slurry Form A 6 2-Me THF 4° C. Evaporation Form A 7 H.sub.2O/MeOH 1/3 Room* Slurry Form A (v/v) Aw~0.5 8 H.sub.2O/MeCN 1/3 4° C. Evaporation Mainly (v/v) Form C Aw~0.9 9 MTBE Room Slurry Form D 10 EtOH Room* Slurry Form A 11 2-Propanol Room* Slurry Form A 12 Nitromethane Room* Slurry Form A 13 H.sub.2O Room Slurry Form A 14 DMSO drops Room Slurry Form B dispersed into water *Clear solution observed initially; continued stirring @ 4° C. for 1 day followed by RT for 9 days

Example 8: Polymorph Screen

(181) Polymorph screening at 50° C. was performed. A solution starting with Form B was prepared in a concentration of about 50 mg in 100 to 300 µL of solvents listed in Table 10. The solution was stirred at 50° C. for 24 h.

(182) TABLE-US-00010 TABLE 10 Polymorph Screening at 50° C. Condition State Result H.sub.2O Slurry Form A MTBE Slurry Form A CPME Clear solution Evaporation clear film only

Example 9: Lyophilization Study

(183) Attempts were made to generate amorphous material by lyophilizing Form B. Form B was lyophilized with a LyoSys 3 freeze dryer, in DPD core area at -40° C. overnight. Results are shown Table 11.

(184) TABLE-US-00011 TABLE 11 Lyophilization attempts. Experiment Form/Solvent Result 1 240 mg Form B in~4:1 H.sub.2O:MeCN Form A 2 240 mg Form B in~3:2 H.sub.2O:MeCN Form B 3 240 mg Form B in~2:1 H.sub.2O:MeCN Form B

Example 10: Form A Solubility Study

(185) Solubility was evaluated at 2 hr and 24 hr for Form A in FaSSIF-v2 and FeSSIF-v2 at 37° C. Approximately 40 mg of Form A sample was added in 2.0 mL of corresponding bio-medium to form a suspension, followed by shaking at 500 rpm at 37° C. After 2 hr and 24 hr, the supernatants were isolated by centrifugation (7000 rpm for 5 min) followed by filtration through 0.45 µm PTFE membrane and were tested by HPLC concentration and pH. The residual solids were characterized by XRPD. As summarized in Table 12, the 24 hr solubility of Form A in FaSSIF-v2 and FeSSIF-v2 was measured to be 0.063 and 1.1 mg/mL, respectively. The XRPD patterns indicated that no form change was observed after solubility evaluation.

(186) TABLE-US-00012 TABLE 12 Summary of solubility evaluation for Form A at 37 ° C. Time point Solubility Medium (hr) (mg/mL) Observation Final Form Final pH FaSSIF (pH 2 0.063 Turbid Form A 6.5 6.5) FaSSIF (pH 24 0.063 Turbid Form A 6.6 6.5) FeSSIF (pH 2 1.0 Turbid Form A 5.9 5.8) FeSSIF (pH 24 1.1 Turbid Form A 6.0 5.8)

Example 11: Accelerated Solid-State Stability Study of Form A

(187) Solid-state stability evaluation (accelerated conditions) was performed on Form A under conditions of 25° C./60% RH and 40° C./75% RH. The detailed procedure utilized for stability evaluation is listed below:

(188) 1) weigh approximately 30 mg of Form A sample into each 3-mL vial;

(189) 2) cover the vial with Parafilm® and poke 15 holes on the Parafilm® with a needle;

(190) 3) put the vial in a stability chamber of 25° C./60% RH or 40° C./75% RH; and

(191) 4) sample for XRPD, TGA, DSC, HPLC purity and assay tests after 2, 4 and 6 weeks and 3 months.

(192) The results of solid-state stability evaluation are summarized in Table 13. The impurities are summarized in Table 14. Based on (RFD, TGA, and DSC characterization results, no significant form change or HPLC purity decrease was observed after 3 months at both conditions. Therefore, Form A sample showed good physical and chemical stability under 25° C./60% RH and 40° C./75% RH conditions for 3 months.

(193) TABLE-US-00013 TABLE 13 Stability results summary. TGA weight loss HPLC Assay vs. Time Crystal (%), up to DSC Purity 5° C. control point Condition form 100° C.) endotherm (area %) (%) Initial — Form A 1.2 110.9 99.86 NA 2 weeks 25° C./60% RH Form A 1.1 110.1 99.84 101.2 40° C./75% RH Form A 1.1 110.5 99.78 98.1 4 weeks 25° C./60% RH Form A 1.7 110.9 99.86 101.4 40° C./75% RH Form A 1.9 110.8 99.86 101.3 6 weeks 25° C./60% RH Form A 0.2 110.4 99.86 99.7 40° C./75% RH Form A 0.8 110.8 99.86 99.2 TGA weight loss HPLC Assay vs. Time Crystal (%), up to DSC Purity 5° C. control Chiral point Condition form 100° C.) endotherm (area %) (%) Purity Initial NA Form A 1.2 110.9 99.86 NA 100* 3 months 25° C./60% RH Form A 0.03 112.6 99.86 97.3 100 40° C./75% RH Form A 0.01 112.9 99.86 96.9 100 *The chiral purity was tested for the starting material after stored at -20° C. for about 3 months

(194) TABLE-US-00014 TABLE 14 % Area of impurities summary in solid-state stability study. % Area Imp.1 API Imp.2 Imp.3 Time (RRT (RRT (RRT (RRT point Condition 0.81) 1.00) 1.04) 1.74) Initial NA 0.09 99.86 — 0.05 25° C./60% RH 0.07 99.84 — 0.09 2 weeks 40° C./75% RH 0.08 99.78 0.07 0.07 25° C./60% RH 0.07 99.86 — 0.07 4 weeks 40° C./75% RH 0.07 99.86 — 0.07 25° C./60% RH 0.07 99.86 — 0.07 6 weeks 40° C./75% RH 0.07 99.86 — 0.07 % Area Time Imp.1 API Imp.3 point Condition (RRT 0.81) (RRT 1.00) (RRT 1.74) Initial NA 0.09 99.86 0.05 3 months 25° C./60% RH 0.09 99.86 0.05 40° C./75% RH 0.09 99.86 0.05 —: <0.05.

Example 12: Solid State Characterization of Form A

(195) A summary of baseline solid state characterization results of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A, including pKa, partition coefficient (log D7.4) measurement and 2-hrs equilibrium solubility in FaSSIF and FeSSIF (version 2) are provided herein. All results obtained are summarized in Table 15. The pKa values were first predicted by Marvin® and then detected by Sirius T3 with the range from 2.0 to 12.0 (in triplicate). Log D7.4 was measured with the solvent system of pH 7.4 phosphate buffer and n-octanol by shake-flask method. Meanwhile, 2-hrs equilibrium solubility of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate in FaSSIF and FeSSIF (version 2) was measured to be 0.12 and 0.080 mg/mL.

(196) TABLE-US-00015 TABLE 15 Summary of pKa, Log D and solubility results. pKa 2-Hrs equilibrium Predicted Measured solubility (mg/mL) pKa value pKa value Log D.sub.7.4 FaSSIF FeSSIF 2.26, 6.67 5.79 3.76 0.12 0.080 pKa values were predicted and detected in the range of 2.0 to 12.0 N/A: no measurement.

Example 13: Solubility of Form A

(197) Approximate solubility of Form A was determined in 20 solvent systems at RT.

Approximately 2 mg of sample was added into a 3-mL glass vial. Solvents in Table 16 were then added stepwise (100 µL per step) into the vials until the solids were dissolved visually or a total volume of 2 mL was reached. Solubility results summarized in Table 16 were used to guide the solvent selection in screening experiment design.

(198) TABLE-US-00016 TABLE 16 Approximate solubility of Form A at RT Solvent Solubility (mg/mL) Solvent Solubility (mg/mL) MeOH S > 23.0 1,4-dioxane S > 22.0 EtOH S > 17.0 DCM S > 19.0 IPA S > 24.0 CHCl₃ S > 19.0 Acetone S > 17.0 ACN S > 24.0 MIBK S > 24.0 Anisole S > 20.0 EtOAc S > 26.0 toluene 4.5 < S < 6.0 IPAC S > 21.0 n-heptane S < 0.9 MTBE 7.7 < S < 11.5 NMP S > 21.0 THF S > 26.0 DMSO S > 20.0 2-MeTHF S > 23.0 H₂O S > 1.0

Example 14: Solubility of Form B

(199) Solubility of Form B was evaluated at 2 hr for Form B in FaSSIF-v2 and FeSSIF-v2 at 37° C. Approximately 10 to 20 mg of Form B sample was added in 0.8 mL of corresponding bio-relevant medium to form a suspension, followed by shaking at 500 rpm at 37° C. After 2 hr, the supernatants were isolated by centrifugation (7200 rpm for 5 min) followed by filtration through 0.45 mm PTFE membrane and were tested by HPLC concentration and pH. The residual solids were characterized by XRPD. As summarized in Table 17, Form B converted to a new form (Form C) after solubility evaluation, while the solubility in FaSSIF-v2 and FeSSIF-v2 was measured to be 0.12 and 0.080 mg/mL, respectively.

(200) TABLE-US-00017 TABLE 17 Summary of solubility evaluation for Form B at 37° C. Time point Solubility Medium (hr) (mg/mL) Observation Final Form Final pH FaSSIF (pH 2 0.12 Turbid Form C 6.6 6.5) FeSSIF (pH 2 0.08 Turbid Form C 5.9 5.8)

(201) Approximately 55 to 90 mg of Form B was added into a 4 mL glass vial. Solvent list in Table 18 was added into each of the vial. These samples were stirred for 3 minutes prior to visual solubility assessment.

(202) TABLE-US-00018 TABLE 18 Summary of solubility evaluation for Form B at room temperature. 1 Ethyl Acetate >800 mg/mL 2 Isopropyl Acetate <810 mg/mL 3 Acetone >430 mg/mL 4 Dichloromethane >830 mg/mL 5 Acetonitrile >720 mg/mL 6 2-Me THF >810 mg/mL 7 H₂O/MeOH 1/3 (v/v) Aw~0.5 >780 mg/mL 8 H₂O/MeCN 1/3 (v/v) Aw~0.9 >730 mg/mL 10 EtOH >800 mg/mL 11 2-Propanol >365 mg/mL 12 Nitromethane >560 mg/mL

Example 15: Anti-solvent Addition Experiments

(203) Anti-solvent addition experiments were performed. A total of 8 anti-solvent addition experiments were carried out. Approximately 15 mg of Form A was dissolved in 0.5 mL solvent to obtain a clear solution, and the solution was magnetically stirred, followed by the addition of the anti-solvent to induce precipitation until the total amount of anti-solvent reached 10 mL. The precipitate was isolated for XRPD analysis. Clear solutions were transferred to -20° C. for 5 hr. Results summarized in Table 19 showed that only Form A was obtained.

(204) TABLE-US-00019 TABLE 19 Summary of anti-solvent addition experiments. Solvent Anti-solvent Solid Form EtOH H₂O Form A* 1,4-dioxane Form A DMSO Form A THF Form A EtOAc Form A CHCl₃ n-heptane Form A* Acetone Form A* THF Form A*: These solids were obtained by slurry at -20° C.

Example 16: Slow Evaporation Experiments

(205) Slow evaporation experiments were performed under 6 conditions. Approximately 15 mg of Form A was dissolved in 1.0 mL of corresponding solvent in a 3-mL glass vial. The visually clear solutions were evaporated at RT to induce precipitation. The solids were isolated for XRPD analysis. Results summarized in Table 20 indicated that Form A, Form B, and amorphous were obtained.

(206) TABLE-US-00020 TABLE 20 Summary of slow evaporation experiments. Solvent (v/v) Solid Form MeOH Form A IPAC Form A 2-MeTHF Amorphous ACN Amorphous EtOH:H₂O (9:1) Form A ACN:H₂O (9:1) Form B

Example 17: Slow Cooling Experiments

(207) Slow cooling experiments were conducted in 2 solvent systems. Approximately 50 mg of Form A was suspended in 0.51.0 mL of solvent in a 3-mL glass vial at RT. The suspension was then heated to 50° C., equilibrated for about 2 hr and filtered into a new vial. Filtrates were slowly cooled down to 5° C. at a rate of 0.1° C./min. Clear solutions were transferred to -20° C., followed

by evaporation to dryness at RT and then solids were tested by XRPD. Results summarized in Table 21 indicated only Form A was obtained.

(208) TABLE-US-00021 TABLE 21 Summary of slow cooling experiments. Solvent (v/v) Solid Form MTBE/EtOAc (9:1) Form A* Toluene/EtOH (9:1) Form A* *: These solids were obtained by evaporation at RT

Example 18: Slurry Conversion Experiments

(209) Slurry conversion experiments were performed. Slurry conversion experiments were conducted at RT in 7 different solvent systems. Approximately 20 mg of Form A was suspended in 0.30.9 mL of solvent in a HPLC glass vial. After the suspensions were stirred for 5 days at RT, the remaining solids were isolated for XRPD analysis. Clear solutions were obtained after stirring, in which anti-solvent of n-heptane was added to induce crystallization. Results summarized in Table 22 indicated that only Form A was obtained.

(210) TABLE-US-00022 TABLE 22 Summary of slurry conversion experiments at RT. Solvent (v/v) Solid Form MTBE Form A Toluene Form A Acetone/n-heptane (1:2) Form A THF/n-heptane (1:2) Form A* 1,4-dioxane/H.sub.2O (1:2) Form A IPA/H.sub.2O (937/63, aw~0.5) Form A H.sub.2O Form A *: Solids were obtained after anti-solvent addition of n-heptane.

(211) Slurry conversion experiments were also conducted at 50° C. in 3 different solvent systems. Approximately 20 mg of Form A was suspended in 0.41.0 mL of solvent in a HPLC glass vial. After the suspensions were stirred for 5 days at 50° C., the remaining solids were isolated for XRPD analysis. Clear solutions were obtained after stirring, in which anti-solvent of n-heptane was added to induce crystallization. Results summarized in Table 23 showed only Form A was obtained.

(212) TABLE-US-00023 TABLE 23 Summary of slurry conversion experiments at 50° C. Solvent (v/v) Solid Form MTBE Form A* Toluene Form A* H.sub.2O Form A *: Solids were obtained after anti-solvent addition of n-heptane.

Example 19: Liquid Vapor Diffusion Experiments

(213) Liquid Vapor Diffusion experiments were performed. Four liquid vapor diffusion experiments were conducted. Approximately 15 mg of Form A was dissolved in 0.5 mL of solvent to obtain a clear solution in a 3-mL vial. This solution was then placed into a 20-mL vial with 4 mL of anti-solvent. The 20-mL vial was sealed with a cap and kept at RT for organic vapor to interact with the solution. The precipitates were isolated for XRPD analysis. The results summarized in Table 24 showed that Form A and Form B were obtained.

(214) TABLE-US-00024 TABLE 24 Summary of liquid vapor diffusion experiments. Solvent Anti-solvent Solid Form THF H.sub.2O Form A EtOH Form A EtOAc Form B CHCl.sub.3 n-heptane Form A

Example 20: Interconversion Study

(215) An interconversion study of crystal forms was performed. To understand the thermodynamic stability relationship between Form A and B, competitive slurry experiments were performed at RT and 50° C. in water and toluene. Prior to the study, Form A was used to saturate the corresponding solvent before it was filtered to obtain a near-saturated solution. Equal amounts of Form A and B samples were weighed and then mixed with the prepared near-saturated solution to form a new suspension, which was stirred magnetically (~1000 rpm) at RT and 50° C. As summarized in Table 25, Form A was obtained after slurry at RT and 50° C. for 1-3 weeks, indicating that Form A was thermodynamically more stable than Form B from RT to 50° C.

(216) TABLE-US-00025 TABLE 25 Thermodynamic stability study. Starting Form Solvent (v/v) Temperature Final Form Form A + B H.sub.2O/PhMe RT Form A Form A + B H.sub.2O/PhMe 50° C. Form A

Example 21: Competitive Slurry Study

(217) A competitive slurry study of all known forms A to E was performed in water at room temperature to understand the thermodynamic stability relationship among these forms. Form A was used to saturate the water before it was filtered to obtain a near-saturated solution. Equal

amounts of each form (A to E) were weighed and then mixed with the prepared near-saturated solution to form a new suspension, which was stirred magnetically (~1000 rpm) at room temperature for 7 days. Solids were isolated and analyzed by XRPD and Form A was obtained indicating that Form A was thermodynamically the most stable form at RT.

Example 22: Vapor Induced Crystallization Experiments

(218) A total of 9 different experiments were conducted. About 100 mg of Form B was placed in a 4 mL glass vial which was then placed into a 20-mL vial with 4 mL of anti-solvent in it. The 20-mL vial was sealed with a cap and kept at RT for organic vapor to interact with the solid. After 7 to 10 days, wet solids were isolated and analyzed by XRPD. The results summarized in Table 26 showed that Form A and Form B were obtained.

(219) TABLE-US-00026 TABLE 26 Summary of vapor induced crystallization experiments

Condition	Result	Ethanol	Form A	DMSO	Form A	Isopropyl	Acetate	Form A	Methyl	Acetate	Form B	THF	Clear film after evaporation	CH ₂ Cl ₂ .sub.2	Form B	Cyclohexanone	Form A	Dimethyl

Formamide Form A N-butyl Acetate Form A

(220) Characterization Methodology

(221) Samples generated as described in the solid form screen were typically analyzed by X-Ray Powder Diffraction (XRPD). XRPD was conducted on a PANalytical Empyrean and X'Pert3 X-ray powder diffractometer using Cu K α radiation at 1.54 Å. In general, positions of XRPD peaks are expected to individually vary on a measurement-by-measurement basis by about $\pm 0.2^\circ$ 2θ . In general, as understood in the art, two XRPD patterns match one another if the characteristic peaks of the first pattern are located at approximately the same positions as the characteristic peaks of the second pattern. As understood in the art, determining whether two XRPD patterns match or whether individual peaks in two XRPD patterns match may require consideration of individual variables and parameters such as, but not limited to, preferred orientation, phase impurities, degree of crystallinity, particle size, variation in diffractometer instrument setup, variation in XRPD data collection parameters, and/or variation in XRPD data processing, among others. The determination of whether two patterns match may be performed by eye and/or by computer analysis. An example of an XRPD pattern collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 1.

(222) PANalytical Emptyrean

(223) Scan Mode: Continuous

(224) Detector: D/teX Ultra

(225) Divergence slit: Automatic

(226) Start Angle: 3 deg.

(227) Stop Angle: 40 deg.

(228) Step Width: 0.0200 deg.

(229) Scan time (s): 5.00 deg./min

(230) Incident Slit: $\frac{1}{2}$ deg.

(231) Receiving Slit #1: 8.000 mm

(232) Receiving Slit #2: 13.000 mm

(233) Wavelength: K α 1: 1.540598 Å ; K α 2: 1.5444426 Å; intensity ratio K α 2/K α 1: 0.50; 40 kV, 44 mA.

(234) X'Pert3

(235) Scan Mode: Continuous

(236) Divergence slit: $\frac{1}{8}^\circ$

(237) Start Angle: 3 deg

(238) Stop Angle: 40 deg.

(239) Step Size: 0.0263 deg.

(240) Scan step time (s): 46.665

(241) Test time: about 5 minutes

(242) Wavelength: $K\alpha 1$: 1.540598 Å; $K\alpha 2$: 1.5444426 Å; intensity ratio $K\alpha 2/K\alpha 1$: 0.50

(243) Differential Scanning calorimetry (DSC) analyses were performed on a TA Instruments Q200/Q2000. About 5 mg of sample was placed into a tared DSC crimped/open aluminum pan and the weight of the sample was accurately recorded. The heating rate was 10° C./min, using nitrogen as the purge gas. An example of a DSC thermogram collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 2.

(244) Thermal Gravimetric Analyses (TGA) were performed on a TA Instruments Q500/Q5000. About 10 mg of sample was placed on an open aluminum pan, accurately weighed and loaded into the TGA furnace. The heating rate was 10° C./min, using nitrogen as the purge gas. An example of a TGA thermogram collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 2.

(245) Dynamic vapor sorption (DVS) was performed on a Surface Measurement Systems DVS Intrinsic. The relative humidity at 25° C. were calibrated against deliquescence point of LiCl, Mg(NO₃)₂, and KCl. Sample sizes were between 10-20 mg. Gas and flow rate was nitrogen, 200 mL/min; dm/dt: 0.002%/min.

(246) HPLC was performed with Agilent 1100/1260. The analysis for Form A was determined with a Phenomenex Gemini C18, 150×4.6 mm, 3 nm (column). The mobile phase was 0.1% formic acid in water; flow rate of 1.0 mL/min; column temperature of 40° C.; and UV detector at 270 nm. The analysis for Form B was determined with a Kinetex 5 μm EVO C18, 150×4.6 mm, 5 μm (column). The mobile phase was 0.1% formic acid in acetonitrile; flow rate of 1.0 mL/min; column temperature of 40° C.; and UV detector at 210 nm.

(247) Proton NMR was collected on Bruker 400 NMR Spectrometer using DMSO-d₆.

(248) The embodiments described above are intended to be merely exemplary, and those skilled in the art will recognize, or will be able to ascertain using no more than routine experimentation, numerous equivalents of specific compounds, materials, and procedures. All such equivalents are considered to be within the scope of the disclosure and are encompassed by the appended claims.

(249) Citation or identification of any reference in this application is not an admission that such reference is available as prior art. The full scope of the disclosure is better understood with reference to the appended claims.

Claims

1. A single unit dosage form comprising a crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, wherein: the crystalline form is Form A of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.1, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ ; the crystalline form is Form B of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees $2\theta \pm 0.2$ degrees 2θ ; the crystalline form is Form C of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees $2\theta \pm 0.2$ degrees 2θ ; the crystalline form is Form D of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees $2\theta \pm 0.2$ degrees 2θ ; or the crystalline form is Form E of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees $2\theta \pm 0.2$ degrees 2θ .
2. The single unit dosage form of claim 1, wherein the crystalline form is Form A of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.1, 20.1, and 22.3 degrees $2\theta \pm 0.2$

degrees 2θ .

3. The single unit dosage form of claim 2, wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.7 degrees $2\theta \pm 0.2$ degrees 2θ .

4. The single unit dosage form of claim 1, wherein the crystalline form is Form B of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees $2\theta \pm 0.2$ degrees 2θ .

5. The single unit dosage form of claim 4, wherein the X-ray powder diffraction pattern further comprises peaks at 15.1, 18.7, and 20.1 degrees $2\theta \pm 0.2$ degrees 2θ .

6. The single unit dosage form of claim 1, wherein the crystalline form is Form C of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees $2\theta \pm 0.2$ degrees 2θ .

7. The single unit dosage form of claim 6, wherein the X-ray powder diffraction pattern further comprises peaks at 12.3, 16.8, and 25.5 degrees $2\theta \pm 0.2$ degrees 2θ .

8. The single unit dosage form of claim 1, wherein the crystalline form is Form D of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees $2\theta \pm 0.2$ degrees 2θ .

9. The single unit dosage form of claim 8, wherein the X-ray powder diffraction pattern further comprises peaks at 18.3, 26.7, and 29.5 degrees $2\theta \pm 0.2$ degrees 2θ .

10. The single unit dosage form of claim 1, wherein the crystalline form is Form E of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees $2\theta \pm 0.2$ degrees 2θ .

11. The single unit dosage form of claim 10, wherein the X-ray powder diffraction pattern further comprises peaks at 15.8, 18.4, and 21.9 degrees $2\theta \pm 0.2$ degrees 2θ .

12. The single unit dosage form of claim 1, wherein the single unit dosage form is suitable for oral, mucosal, rectal, nasal, vaginal, parenteral, subcutaneous, intramuscular, bolus injection, intraarterial, intravenous, sublingual, transdermal, buccal, or topical administration.

13. The single unit dosage form of claim 12, wherein the single unit dosage form is a tablet, caplet, capsule, powder, or sterile solid that can be reconstituted to provide a liquid dosage form.

14. The single unit dosage form of claim 12, wherein the single unit dosage form is a tablet, caplet, capsule, or liquid suitable for oral administration.

15. The single unit dosage form of claim 12, wherein the single unit dosage form is a solution ready for injection, dry product ready to be dissolved or suspended in a pharmaceutically acceptable vehicle for injection, suspension ready for injection, or an emulsion ready for injection.

16. The single unit dosage form of claim 1 comprising about 0.1 mg to about 1,000 mg of the crystalline form.
